

Desktop multimedia

T Midwinter

In the last few years the technology to deliver conversational multimedia services to the desktop has begun to mature. At the same time, the international standards required to ensure seamless interoperability have been ratified, leading to the ready availability of systems to convert a mid-range PC into a desktop multimedia system. As the market begins to settle, a wide range of products and services will become available allowing businesses and individuals to live and work in ways they had never believed possible. Physical distance will no longer be a barrier to business relationships.

This paper looks at some of the technology and standards that are delivering this vision. It starts with a look at the anatomy of a desktop system, progresses to looking at the standards on which it is based, and finishes with a discussion of groupworking in a real-world environment.

1. Introduction

Multimedia has become a commonly used term within the computer industry. Every week there seems to be a new multimedia product launched by one computer company or another. The term 'multimedia' in the context of these products has become synonymous with CD encyclopaedias and stunning graphics performance, but has had little real impact on most businesses. 'Multimedia communications', on the other hand, is a very new topic that has just started to be addressed by the computer and communications industries. Multimedia communications promises to radically change the way business operates and in the longer term is likely to have as major an effect on the fabric of our society as the Industrial Revolution and the invention of the telephone.

Desktop multimedia is rapidly becoming the latest term to escape from the computer industry's marketing department. It is a term used to describe the amalgamation of 'multimedia communications' with the PC on a user's desk. The resultant desktop multimedia terminal consists of a standard PC (configured for the user's normal business) with additional functionality to allow it to communicate with a telephone network. It will allow the user to make calls to anybody around the world, using existing telephone networks. If the other party in the call is capable of video and/or data communications, then the terminal will automatically set up a call to the full capabilities of the two ends, without any further user intervention. One terminal on the desk will provide all the communications needs of the user, allowing the existing telephone, facsimile machine and data modem to be thrown away. One line, one terminal and one unified service — that is the 'multimedia

communications' vision, and, as will be shown in this paper, it is well on the way to becoming reality.

Desktop multimedia is much more than just a technology to allow users to connect their PCs to the telephone network and transmit speech and video at the same time as data. It is an enabling technology that will allow businesses much more flexibility in balancing the needs of their customers and workforce. The customer needs are better met as the business can call on specialists when required. The workforce needs are better met as people will have much more choice regarding where they live, and what hours they work. In essence, the technology promises to break down many of the barriers that distances (and hence travel time) currently impose. If you can ring any colleagues at will, speak to them, see them and discuss, modify or transfer any document or other data with them, then travelling into the office becomes much less necessary. If all these functions can be performed within a multiparty conference, then many people will be able to work from home rather than having to commute. Now include the problems faced by multinational companies, as they try to get international teams of experts together to discuss an issue, and the advantages of working together efficiently, despite the physical distances, become obvious. Desktop multimedia provides one of the base technologies required for all of these work styles — consider the following job example.

- Telephone operator [1] — modern digital telephone systems will allow calls to be answered as easily using a terminal at home as using a terminal in a central office. The telephone system can present a call to the

operator's home number and hence the job could be done as efficiently from home as from a central office. The advantages are obvious — zero travelling time, easier emergency cover provision, less centralised accommodation, etc. The major disadvantage would seem to be that the operator could become rather isolated from colleagues, and lose the feeling of company loyalty. However, trials have proved that the use of video as well as audio in conversations allows teleworkers to interact as part of a tightly knit community [2]. Although this inevitably incurs extra call charges, the benefits easily outweigh the costs.

Although the example chosen is well understood and documented, it can easily be extended to include most other office-based jobs.

Desktop multimedia has recently become possible due to the merging of many development strands. As well as the technology, standards have played a key role. Over the last ten years, standards for videoconferencing and multimedia have been developed to achieve the vision of global interconnect, at the basic capability level of both terminals. These standards, together with the remorseless march of technology, are now beginning to deliver the dream — one terminal connected via one telephone line across the world-wide telephone system to deliver a simple-to-use, consistent service from telephony at one end of the scale to desktop multimedia at the other (see Fig 1).

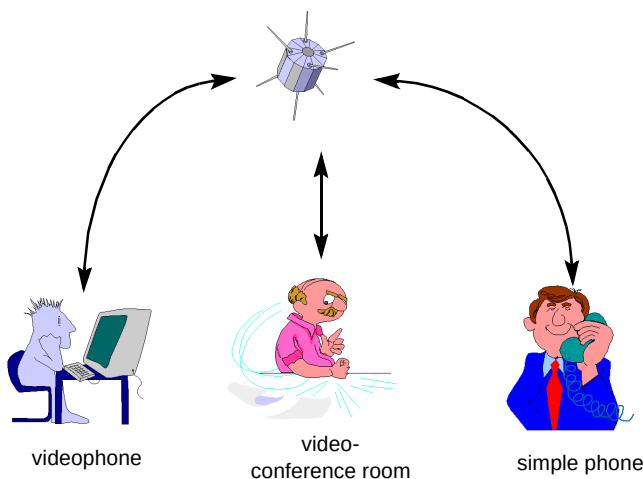


Fig 1 World-wide interconnect

2. A generic desktop multimedia system

A desktop multimedia system can be considered as two items, all the add-ons to a PC and the PC itself. This paper will look at a general model of a desktop multimedia system that is capable of connection to an ISDN basic rate interface (ISDN BRI [3]) network (see Fig 2).

If a desktop multimedia system is to replace the user's existing telephone, it must be capable of operating like a standard telephone, even with the PC switched off, as well as a full multimedia videophone when the PC is running. Similarly, once powered, the PC should be capable of communicating with other data devices, like facsimile and modem systems. The functionality for these is given below.

2.1 Basic telephony functionality

To perform as a simple telephone, when the PC is not powered, puts considerable limitations on the audio unit (telephone) of a desktop multimedia system. However, if the user's normal phone is to be substituted by the desktop product then these limitations must be overcome. The major features are:

- the audio unit must operate even when there is no power to the PC, i.e. all the basic telephone functionality must be line powered or powered from a plug-top PSU,
- the audio unit must be capable of originating and answering calls,
- the audio unit must be capable of operating in an office environment, i.e. a handset is required — in some cases loudspeaking and headset operation would also be useful,
- the audio unit must also be capable of being controlled directly by the PC, when it is powered up, to allow calls to be set up and answered by the PC directly.

Hence the audio unit looks similar to a telephone and in a simple videophone application could be a standard telephone. However, in a desktop multimedia application the telephone needs to be tightly integrated with the PC, to allow the PC and phone to interact during call set-up, call answer, etc; consequently, the audio unit is usually a custom unit designed specifically for the desktop multimedia system. The audio unit needs to include the ISDN BRI interface.

2.2 Videophone-specific functionality

To perform as a standards-compliant videophone, the system needs to be able to compress and decompress both

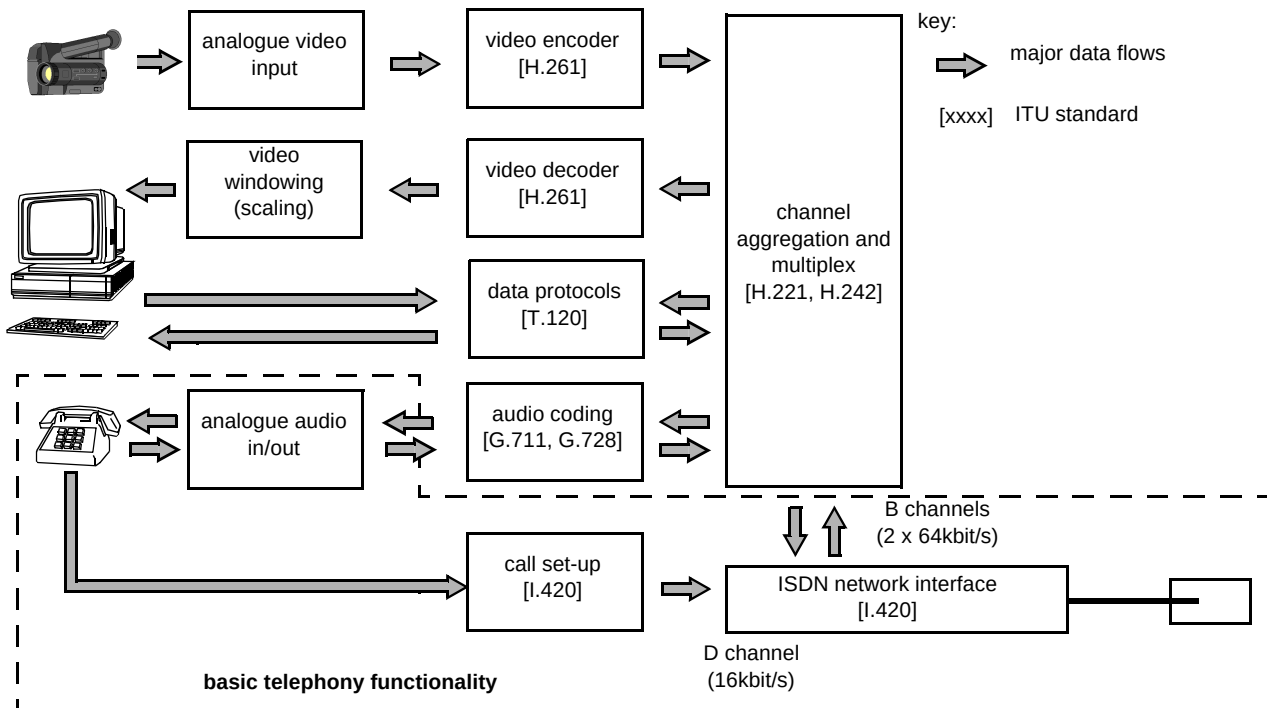


Fig 2 Generic design of a desktop multimedia system.

the video and speech signals. It also needs to be able to aggregate the two 'B' channels presented by the ISDN BRI together to make a single 128-kbit/s data stream. All of these functions are governed by standards, which will be discussed briefly here and in more depth in the standards section of this paper. The major features of the videophone-specific functionality are:

- video encode and decode to ITU-T Recommendation H.261 [4],
- audio encode and decode to ITU-T Recommendation G.711 [5] (Standard Telephone Quality 'A'-Law/'μ'-Law Coding),
- optional audio coding to ITU-T Recommendation G.728 [6] (Low Bit Rate, Telephone Quality Coding) and/or ITU-T Recommendation G.722 [7] (Medium Wave Radio Quality Coding),
- channel aggregation and multiplex to ITU-T Recommendation H.221 [8] and ITU-T Recommendation H.242 [9] (aggregation of the two 64-kbit/s channels together to make a single 128-kbit/s channel and multiplex audio, video and data together into that single channel).

The videophone specific hardware in a desktop multimedia system can be powered from the PC as it does not need to operate when the PC is not operational.

2.3 Desktop multimedia-specific functionality

The differences between a desktop multimedia system and a standard videophone are based around the PC integration. The major features of the desktop multimedia specific functionality are:

- PC application software to control all the telephony functions, video windowing, facsimile, modem and any other functionality provided,
- PC application software to transfer data from one party to another using the ITU-T Recommendation T.120 [10] protocols — these applications include file transfer, shared whiteboard, application sharing, multipoint conference control, network database control, etc (this data will need to be inserted into the data channel for transfer to the remote party(ies)),
- facsimile and modem algorithms,
- video windowing hardware may need to be added if the host PC does not include it.

2.4 The complete desktop multimedia system

When all the modules described above, and a camera, are integrated together, a complete desktop multimedia system is produced.

The system discussed so far has not made any distinction between add-on functionality, and software running on the host itself. This is deliberate, as the split between host and accelerator card is continually changing. The functions listed above are generic and apply to all systems. Their implementation is product specific, creating the cost/performance differentiation between manufacturers.

To highlight the differences, two products are considered — a hardware accelerator product designed to run on most Windows 3.1-capable PCs, and a software-only product which requires a top-end PC.

- Case study 1— the BT VC8000 multimedia videophone (using hardware acceleration)

The VC8000 is in many ways the traditional videophone design, in that it does not require much support from the host PC. All of the functions described for the generic videophone are performed on the accelerator card, leaving the PC free to run Windows, user applications and multimedia applications. The processing power available on the accelerator card is approximately three times that of a top-end Pentium processor, which allows the card to produce good-quality video pictures, implement all of the audio-coding algorithms (including the computationally intensive G.728 [6]) and provide T.120 [10] and ISDN interface support.

The hardware provided in a VC8000-type kit includes a telephone (which operates with the PC power off), a camera, an ISDN interface, a windowing interface and all the processing to implement the video, audio and data coding. Irrespective of whether a call is in progress, the processing power of the host PC is not compromised by the operation of the videophone.

Hardware acceleration tends to offer the customer a high-quality product that makes far fewer demands on the host PC than the 'software only' solutions. The cost of this type of solution is also greater than a 'software only' solution, but the differential can be much smaller than expected as even the 'software only' system requires some added hardware.

- Case study 2 — a software-based system (no hardware acceleration)

A 'software only' codec may initially appear to be a very cheap solution, but is, in practice, still quite expensive as it requires hardware which includes a camera, an audio sub-system and an ISDN interface. Some of these features are currently provided on top-end multimedia PCs, but it is not clear that they are really suitable for a full telephony-quality product. In a noisy office, would a loudspeaking PC really be acceptable in place of a standard telephone?

The software codecs developed so far have relied on an ISDN card, which also provides PCM audio to the PC and a digitising card, with camera, to provide a video source. The H.261 [4], H.242 [9] and H.221 [8] modules are performed by the PC in real time. The PCs used for this type of codec are top-end 486s (66 MHz) or Pentiums, and the videophone operation takes up most of the processing power of the PC. Applications run slowly, or the video stops when both video and applications are active at the same time.

Another major problem with all software-based codecs is that they require the PC to be switched on even to operate as a basic telephone. Consequently, if the PC is switched off, or the application shut down, the ISDN telephone will effectively be unplugged and calls cannot be made or received — not a real videophone solution. In the future there will be phone add-on kits for PCs which will allow basic telephony even with the PC unpowered.

3. International standards and interworking

The ITU-T, previously known as the CCITT, has been responsible for setting telecommunications standards for many years. During this time it has produced recommendations for all forms of communications equipment ranging from basic telephones, facsimiles and modems up to trunk signalling equipment and telephone exchanges. In conjunction with PTTs and manufacturers from around the world the ITU-T has also defined sets of standards for desktop multimedia systems. The umbrella standard for multimedia terminals on the ISDN is H.320 [11]. For multimedia terminals on the PSTN, the recommendation, to be known as H.324, is in the final stages of agreement. One of the features of ITU-T standards is that they take a considerable amount of effort to get correct, but once ratified they can be upgraded, while still maintaining backwards compatibility. This leads to families of products from different manufacturers which interwork with all versions of the standards and removes much of the very costly 'designed for obsolescence' associated with most of today's 'Industry Standards'. The major components in H.320 [11] are given below.

3.1 H.261 — the video coding standard

H.261 [4] is the ITU-T standard for real time video coding and defines three main areas within the video coding process.

- Video format

The video format chosen proved to be a very contentious issue when H.261 [4] was developed, as the European requirement was for a PAL-based format and the American/Japanese preference was an NTSC-based format. The final format was a compromise

which was acceptable to all countries and is the Common Intermediate Format (CIF). There has also been a defined lower resolution video format called Quarter CIF (QCIF). As all H.261-based systems use the same video formats, they all interwork together irrespective of whether the originating country operates to the PAL or NTSC TV standard.

- Video decoder

The standard defines how a reference decoder will work, but does not completely define how an encoder works. At first sight this might seem a bit strange, but is actually very important, as it allows manufacturers considerable flexibility in the encoder design and control. This allows each manufacturer to develop a unique encoder design, thus providing much needed product differentiation between systems.

- Error correction

A BCH (511,493) error corrector is defined in the standard. This corrector takes 493 bits of data and adds 18 correction bits, to make up a 511-bit frame, for transmission to the far end. The decoding terminal strips out the correction bits and then has three choices, do nothing (throw the bits away), implement a burst corrector which corrects any single burst of up to 6 bits in the 511-bit frame, or implement a random error corrector which can correct any two errors in the frame. Most real systems opt for one of the two correction schemes. Which scheme is best depends on the actual error characteristics of the line, and, in practice, there seems to be little to choose between the two schemes.

3.2 G.711, G.722 and G.728 — the audio coding standards

Three different audio standards are defined within H.320 [11].

- Telephony quality speech, 'A'-Law and 'μ'-Law

This standard is mandatory and all terminals which claim H.320 [11] compliance must include at least G.711 [5] support. However, this is not very onerous, as G.711 [5] is the same standard as is used to define normal telephony-quality speech. In Europe, the 'A'-Law version of the standard is used and in the US and Japan the 'μ'-Law version is used. G.711 [5] supplies 3.4-kHz analogue bandwidth using 64 kbit/s of channel capacity.

- Low bit rate speech

This standard is optional, although most serious terminals include it. The major exceptions are the software (PC) codecs which do not have the processing power to perform G.728 [6]. This standard gives 3.4-kHz analogue bandwidth (telephony quality) using 16 kbit/s of channel capacity. Using G.728 [6] it is quite practical to conduct a multimedia call including speech, video and data over a single 64-kbit/s ISDN channel.

- Wideband audio

This standard is also optional, and is the one favoured by high-quality speech terminals. G.722 [7] supplies 7-kHz analogue bandwidth (approximately medium-wave radio quality) using 56 kbit/s or 48 kbit/s of channel bandwidth.

Videophone terminals always begin a connection using the G.711 [5] standard as this allows them to instantaneously communicate with all other H.320 [11] terminals as well as normal telephones. Most systems then attempt to negotiate to move to either G.728 [6] to maximise the channel available for video and data, or G.722 [7] to improve the audio quality.

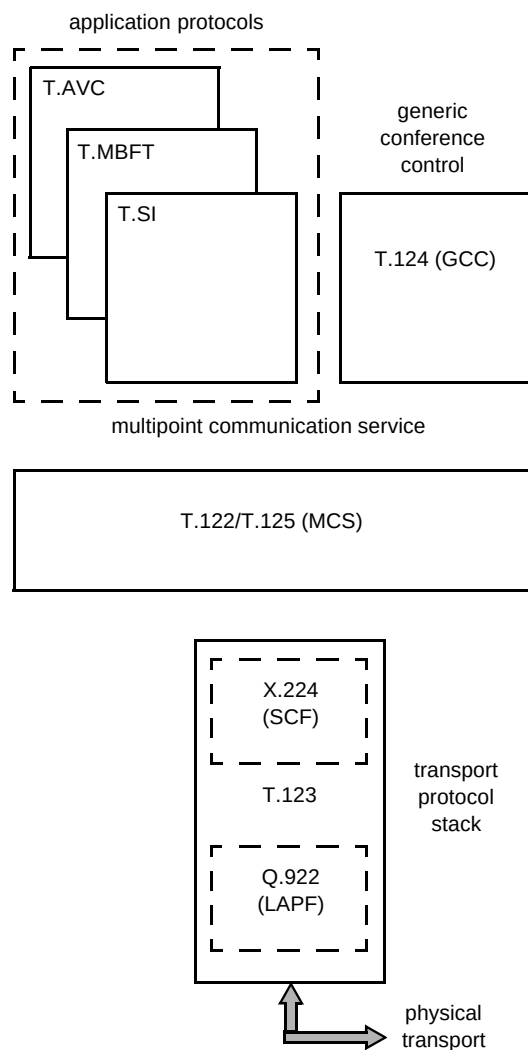
3.3 H.221 and H.242 — the channel aggregation and multiplex standards

H.221 [8] and H.242 [4] perform the following roles.

- H.221 [8] provides facilities to allow multiple 'B' channels to be aggregated together. This allows any differential delay between the two channels in a 2B ISDN call to be removed, creating a single 128-kbit/s channel rather than two 64-kbit/s channels.
- H.242 [9] provides facilities to allow a transmitting terminal to determine the capabilities of the receiving terminal. This allows terminals to automatically disable functions which are not supported by the other terminal. For example, if the far terminal states that it does not support G.728 [6] audio, the sending terminal will not use it during the call.
- H.242 [9] and H.221 [8] provide facilities for defining and signalling the multiplex structure to the remote end so that it is capable of identifying and extracting the individual audio/video/data components. This multiplex structure can be changed every 20 ms during a call, which allows the trade-off between audio, video and data to be dynamically varied to meet the user's requirements.

3.4 T.120 — the data standard

T.120 [10] is an umbrella standard for a series of standards which together define the protocols required for collaborative working (see Fig 3). The T.120 [10] standards are layered, covering all aspects of the protocol from the transport layer to the conferencing layers and finally the shared application layers. Although T.120 [10] is the last of the multimedia standards to be finalised, it has now attracted a sufficient international following to ensure that it remains the data conferencing standard. Microsoft have recently announced that they will be including T.120 [10] in future versions of Windows 95 and Windows NT.



Notes:

T.AVC = audio video control

T.MBFT = multipoint binary file transfer

T.SI = still image

- The transport stack T.123 [12]

T.120 [10] has been designed to provide multipoint data services across a wide variety of data networks such as PSTN, ISDN, LAN and B-ISDN. The transport protocol stack is the layer where the network dependence is removed. The top of the transport layer provides four prioritised, error-free channels between the two ends. The software above the transport stack can assume an error-free transmission of the data submitted to the channel. The four channels of differing priority are provided to allow real time data (e.g. cursor movements) to pre-empt background data transfers (e.g. file transfers). The transport protocol stack is changed depending on the target network. For the ISDN the lower levels of the Q.922 protocol result in synchronous (HDLC) packets being presented to the network. For the PSTN the lower levels of the Q.922 protocol implement an asynchronous (RS232-type) protocol.

- The multipoint communications service (T.122 [13]/T.125 [14])

The multipoint communications service (MCS) provides a link layer which allows any terminal within a conference to communicate with any other terminal(s) as required. Each pipe (link) is named and connects a defined set of terminals together, with a defined priority of transport. Hence, if a conference consisted of five terminals A, B, C, D and E, then MCS provides the capabilities for A, B and E to set up a pipe between them for a private, background, file transfer while at the same time operating a pipe between all five terminals for a multipoint shared application. MCS supports up to 64K simultaneous connections within a call, which should be enough for any requirement.

Hence, MCS can be viewed as a telephone network which allows point-to-point and multipoint calls between the members of a conference. Members can set up as many different point-to-point and multipoint calls as they require, to allow only some sub-groups within the conference to see some data sets.

- The generic conference control (T.124 [15])

The generic conference control (GCC) module manages the setting up and tearing down of pipes through the MCS. It receives requests for service directly from user applications as well as from the application protocols. The GCC is also responsible for the management of a multipoint conference including chairperson control and reservation functions.

- The application protocols — still image (T.126/T.SI [16])

The still image application protocol is probably the most important protocol to be developed. It defines the way in which still image and annotation information will be shared within a collaborative environment. T.SI [16] provides protocols to enable still image transfer (facsimile), shared annotation (whiteboard/chalkboard) and simple remote control applications (application sharing). These protocols allow manufacturers to connect simple and complex applications together, with all applications being able to operate in a compatible manner. While some manufacturers may define extensions to the basic still-image protocol, they will need to provide an automatic fall-back position for standards-compliant communications with other terminals.

- The application protocols — multipoint binary file transfer (T.127/T.MBFT [17])

The multipoint binary file transfer protocol is the basic file transfer protocol built into T.120 [10]. It includes all the functionality required to operate in a multipoint environment as well as the basic point-to-point transfers. As its name suggests, T.MBFT [17] is designed for the transfer of binary data as well as normal ASCII text. Hence any file on any PC in a conference can be transferred to any number of the other PCs in the call.

- The applications protocols — audio/video control (T.AVC [18])

The audio/video control protocol provides the facilities to manage audio and video data streams outside T.120 [10]. Consequently, in an H.320 [11] terminal, the T.AVC [18] protocol would provide remote control of cameras (pan, zoom, document versus face-to-face), control of the switching within a multipoint control unit (MCU), and any other control facilities required within a multimedia call. T.AVC [18] does not carry any real time audio or video information. It is purely a control channel to provide sophisticated control for the real time audio and video streams within a call.

This section has given an overview to the T.120 [10] standard, describing the work of the ITU-T in defining the transmission protocols associated with shared working. While this work is the minimum required to achieve interworking across the world, the computer industry would like further standardisation to occur to allow applications from many vendors to run on any T.120 [10] protocol stack. This work requires that an API be defined at each layer in the protocol stack. This API work is being performed by an industry group known as the IMTC, which includes representation from PTTs, product manufacturers and of course the computer industry. Microsoft, Intel, BT and

many others are working together to define the APIs which will allow a chalkboard application from one vendor to work seamlessly with a T.120 [10] protocol stack from another.

4. Multiparty conferencing

The discussion so far has concentrated on the design of the terminal equipment required for desktop multimedia. However, another very important area of development is the technology within the network to allow effective group working between desktop multimedia terminals worldwide. This group working is known as ‘multiparty conferencing’ and a network system that provides support for this type of operation is known as a ‘multipoint control unit’ (MCU).

The goal of multiparty operation is to allow users with desktop multimedia terminals, designed to the standards described in the previous section, to interwork together as if they were physically co-located. Existing MCU equipment supports voice bridging and video switching. Future systems will include much more sophisticated services that will be controlled by the T.120 [10] standards described in the previous section.

- Audio within a multiparty call

The audio technology for multipoint operation is very similar to the technology provided in a traditional audio-conferencing bridge. The multipoint unit connects all users together by mixing the audio signals from everyone who speaks and transmitting the mixed signal back to all participants. In a multimedia conference environment the speech paths can also be controlled by the T.120 [10] data path so that one participant (usually the chairperson) can control who may speak. As the technology matures, other control paradigms will be developed to assist in the control of large meetings.

Looking to the future, it is likely that new standards will allow the transmission of stereo imaging to the participants in a conference, helping to create a feeling of audio presence. Each speaker’s voice will be imaged to a virtual location at the receiving terminal, making it easier to identify the speaker in a large conference.

- Video within a multiparty call

The video technology for multipoint operation is very different to the audio technology. Unlike audio, it is not possible to mix the pictures of different speakers together and produce any meaningful results. Hence the video technology relies on switching the video picture to look at the current speaker. This is the technology that is currently available within videoconference MCUs. The newest MCUs are now

beginning to include the chairperson control facilities from T.120 [10], which allow one participant to take control of the conference.

Research into the future of video within multiparty conferences is concentrating on ways to transmit multiple video streams to a single terminal. These multiple streams would allow the viewing of two or more participants at the same time, although the actual quality of each image would depend on the limitations of the channel bandwidth available. As these more sophisticated video systems become available, then it will become necessary to develop mechanisms to control them. These 'continuous presence' video systems will allow users to monitor the reactions of selected other participants, while still focusing on the speaker.

- Data within a multiparty call

The T.120 [10] suite of standards, described in section 3 above, is designed to operate in multipoint as well as point-to-point environments. In fact, the standards include two protocols, T.GCC [15] and T.AVC [18], which are specifically designed to control a multipoint conference. All the collaborative working applications have been developed with multiparty conferencing in mind. This leads to chalkboard-type applications that can share images across a conference, or application-sharing applications that will allow all the participants of a conference to jointly author a document. These facilities and others, like the ability to broadcast a file to some or all conference participants, will radically change the way people view electronic meetings.

When the audio, video and data technologies described above are integrated in the next generation MCUs, the user will be presented with a sophisticated, collaborative working environment, which will remove many of the barriers currently imposed by distance. Instead of jumping on to a plane to attend a meeting, the user will be able to dial up access, see all the participants and jointly work with them in a virtual electronic meeting environment. Initially systems will struggle to provide a scenario as good as an existing meeting, but, as users become more familiar with the tools and the systems become more sophisticated, the environments will expand to provide the facilities of the best designed meeting rooms directly to a user's desktop.

5. Desktop multimedia terminals in the real world

The previous sections of this paper have explored the make-up of a desktop multimedia system. This section of the paper will look at how one of these terminals operates in a world environment, which is populated with terminals of many different types, from different manufacturers.

Initial trials with videophones and desktop videophones have highlighted a number of 'telephony' related requirements for this type of equipment. In particular, while the manufacturers of these terminals have tended to think of them as sophisticated meeting aids, the customers view them as sophisticated telephones. This implies that the desktop terminal needs to behave like a telephone, as well as providing the functionality of a multimedia terminal. How do telephones behave? Some of the features which most managers take for granted are:

- the ability to route all incoming calls via a secretary,
- having only one phone on the desk — if it is in use then incoming calls are routed to the secretary.

These points are very important if users are going to migrate from existing telephones to using desktop multimedia systems. If users keep their standard phones and use desktop systems as well, then the latter will quickly be relegated to second place and will only be used for previously planned, structured meetings, missing much of its potential use.

So it would seem that for a desktop system to be truly successful it needs to meet the telephony needs of the end user as well as the multimedia needs. However, the structure of the products and standards described in the previous sections have been designed with these needs in mind.

5.1 Simple telephony

To provide a basic telephony service is actually quite a challenging requirement for a multimedia system. Throughout Europe the basic telephony service is regulated to ensure that all terminals meet the performance requirements defined in the NET33 [19] standard. If a user's phone is to be replaced with a desktop system, then the desktop system will also need to conform to the requirements of NET33 [19]. However, users demand more from their basic phone than they do from their PC. They do not expect their PC to work until they have switched it on whereas the phone is expected to work all the time. Similarly, they expect their PCs to crash occasionally, or even often, whereas they do not expect their phone to crash. These requirements lead to the conclusion that the PC and the phone should be separated, which implies that the desktop multimedia system should be hosted from a telephone-like instrument rather than trying to replace it.

5.2 PABX functionality

To mount a desktop multimedia product behind a PABX requires that the desktop product is capable of all the PABX

functions available on a normal telephone. This can put considerable demands on the desktop product, particularly if these PABX functions are also expected to work with multimedia calls. However, the desktop system also has advantages over the simple phone, the most obvious being that it has the PC display available to represent some of the more complex PABX facilities. As desktop systems become fully integrated into the telephony environment, users are likely to see much more user-friendly PABX/telephony interfaces becoming available, with the simple facilities available via both PC and telephone handset interfaces and the more complex facilities available via the PC interface only.

5.3 Interworking

As far as the user is concerned, the desktop multimedia terminal is just a phone with more functionality. If the user calls a normal phone, mobile phone or any other existing telephony service, the desktop system is expected to connect directly and behave just like an ordinary phone. However, if the user calls an endpoint which is capable of multimedia functionality, then the user expects the terminal to automatically enable as much functionality as is available to both ends. This process of starting at the lowest level functionality (simple telephony) and building the call up to the highest level supported by both ends is automatically carried out using the standards H.221 [8] and H.242 [9]. As time progresses and more standards become available for multimedia over the PSTN, ISDN and other networks, terminals will have to become more sophisticated to maintain this total interworking capability. Without full interworking, users will quickly get very frustrated as calls start failing for unexplained reasons. Interworking remains one of the major challenges of the multimedia age.

6. Conclusions

Looking to the future, it is clear that desktop multimedia is going to dramatically change our lives. This paper has looked at the technology behind the desktop multimedia equipment which is currently being deployed in the market place and discussed some of the issues which are still being addressed. It has shown that ISDN-based products are now available now that can significantly improve the communications within many companies, leading to enhanced team working and major productivity increases.

Over the next few years, products will become available which operate over the existing LAN-based networks and the existing PSTN telephone network. The performance of these systems is still to be proven and while there is little

doubt that PSTN-based products will become very popular, the telephony performance of a LAN-based product is still to be proven. The success or otherwise of all of these products will ultimately depend on the ease of use and functionality provided.

As time progresses, the number of variants of a terminal giving different price/performance trade-offs will increase and the challenge for the industry will be to contain these systems to ensure continued interoperability. In the next decade there will undoubtedly be an explosion in the penetration of multimedia services, leading to radical changes in the way that people do business.

Appendix

Glossary

ITU-T

International Telecommunication Union — a Geneva based organisation which brings the world's PTTs and other interested parties together to define the standards for telecommunications around the world (previously known as the CCITT).

Desktop multimedia

Multimedia communications integrated into a user's desktop terminal (PC or workstation) to allow full video telephony and multimedia communications functionality from the single terminal.

G.711 [5]

The international (ITU-T) standard for telephony quality audio coding. This standard supports two incompatible modes, 'A'-Law which is used throughout Europe and 'μ'-Law which is used in the US and Japan (audio bandwidth: 3.4 kHz, digital data rate: 64 kbit/s, 56 kbit/s).

G.722 [7]

The international (ITU-T) standard for high-quality audio coding. This standard supports medium-wave radio quality audio (audio bandwidth: 7 kHz, digital data rate: 64 kbit/s, 56 kbit/s and 48 kbit/s).

G.728 [6]

The international (ITU-T) standard for low bit rate audio coding. This standard supports G.711 [5] quality audio at a quarter of the data rate (audio bandwidth: 3.4 kHz, digital data rate: 16 kbit/s).

H.221 [8]

The international (ITU-T) standard for the aggregation of multiple 64 kbit/s channels together, and the multiplexing of audio, video and data on to the aggregated channel.

H.242 [9]

The international (ITU-T) standard for the negotiation of capabilities between terminals to ensure that one terminal does not try to perform an operation which is outside the capability of the other terminal.

H.261 [4]

The international (ITU-T) standard for video coding within a conversational audiovisual call.

H.320 [11]

The international (ITU-T) umbrella standard which covers multimedia communications across the ISDN. This standard calls up the other G, H, I and T series standards listed here.

I.420 [3]

The international (ITU-T) standard for ISDN BRI.

ISDN BRI

ISDN basic rate interface — the standard digital telephone interface used throughout the world, conforming to I.420 [3]. This interface presents two 64-kbit/s bearer 'B' channels and one 16-kbit/s data 'D' channel. The 'B' channels are used to connect together the parties in a call and provide either 64 kbit/s or 128 kbit/s of channel capacity between the desktop multimedia systems. The 'D' channel is used to exchange signalling information with the telephone network for call set-up, etc.

Multimedia

Computer industry term to describe the merging of audio and video with data. The audio and video involved with standard multimedia applications is usually stored locally on the PC or workstation and played back at the same time as the associated data is accessed.

Multimedia communications

Communication industry term to describe the merging of real time audio and video with all forms of data. The audio and video involved in multimedia communications applications are sourced in real time from the remote party(ies) in the call. The data element can include any data that is resident on a PC, workstation or other digital device (e.g. Group 4 Facsimile).

PC

Personal computer — this term has been used in this paper to include IBM compatible PCs, Apple Macintosh PCs and workstations of all types. It covers any computing system that might reside on a user's desk.

T.120 [10]

An umbrella standard for a series of standards which define the protocols for collaborative data working between two or more computers in a conference.

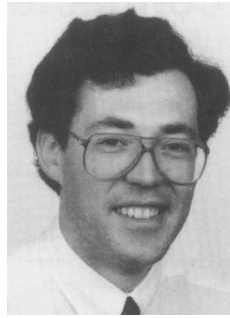
Videophone

A terminal whose major purpose is to provide real time audio and video functionality. Videophones often include a data port which allows them to be connected to a PC or workstation to provide multimedia communications functionality as well.

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